

CLASSIFICATION

CLASSIFICATION TECHNIQUES – PART II

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CLASSIFICATION TECHNIQUES

The following concepts will be introduced:

- ✓ **SUPPORT VECTOR MACHINE**
- ✓ **MARGIN**
- ✓ **LINEAR DECISION BOUNDARY**
- ✓ **LINEARLY SEPARABLE DATA SET**
- ✓ **FEATURE SPACE**
- ✓ **KERNEL FUNCTION**

A **CLASSIFICATION TECHNIQUE (CLASSIFIER)** is a systematic approach to building **CLASSIFICATION MODELS** from a **DATA SET**.

CLASSIFICATION TECHNIQUES can be grouped into:

HEURISTIC; simply yet widely used like DECISION TREES, RANDOM FORESTS, NEAREST NEIGHBOR, ...

REGRESSION BASED; use a parametric conditional probability, LOGISTIC REGRESSION

SEPARATION; partition the attributes' space, **SUPPORT VECTOR MACHINES** and ARTIFICIAL NEURAL NETWORKS

PROBABILISTIC; exploit Bayes formula and compute posterior probability to classify records, NAÏVE BAYES, TREE-AUGMENTED NAÏVE BAYES, ...

We provide a brief overview of some of this techniques even if a formal and detailed treatment of them is out of the scope of this course.

Excellent courses on these techniques are available in Machine Learning MOOC of Andrew Ng.

Given a data set consisting of “m” records where two continuous attributes are measured

$$D = \{(\underline{x}_1, y_1), \dots, (\underline{x}_m, y_m)\}$$

where

$$\underline{x}_i \in \mathbb{R}^2$$

$$y_i \in \{-1, +1\}$$

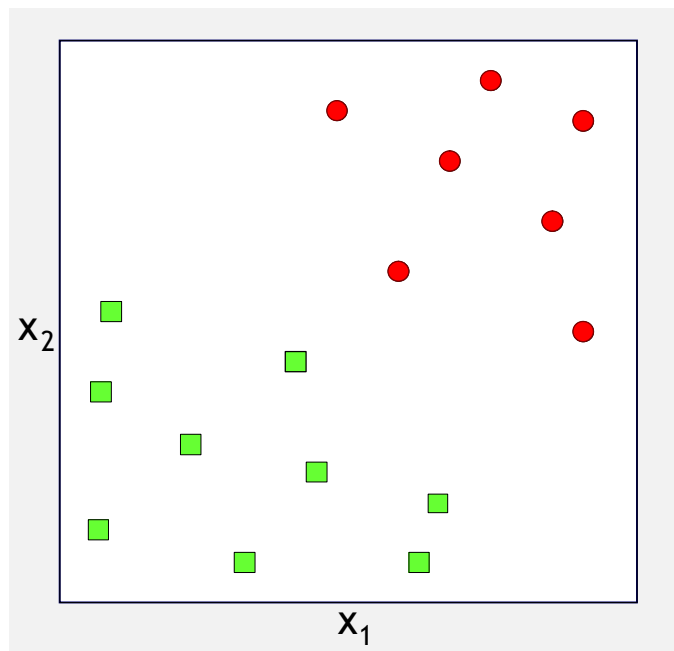
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Is it possible to separate red circles ($y=+1$) from green squares ($y=-1$) ?

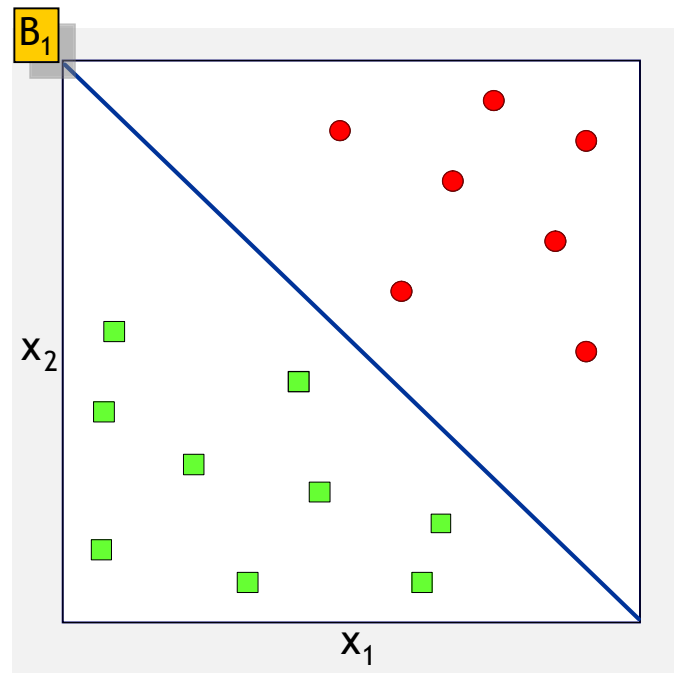
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A possible answer is represented by the blue line, described by the following equation

$$\underline{w} \cdot \underline{x} + b = w_1 x_1 + w_2 x_2 + b = 0$$



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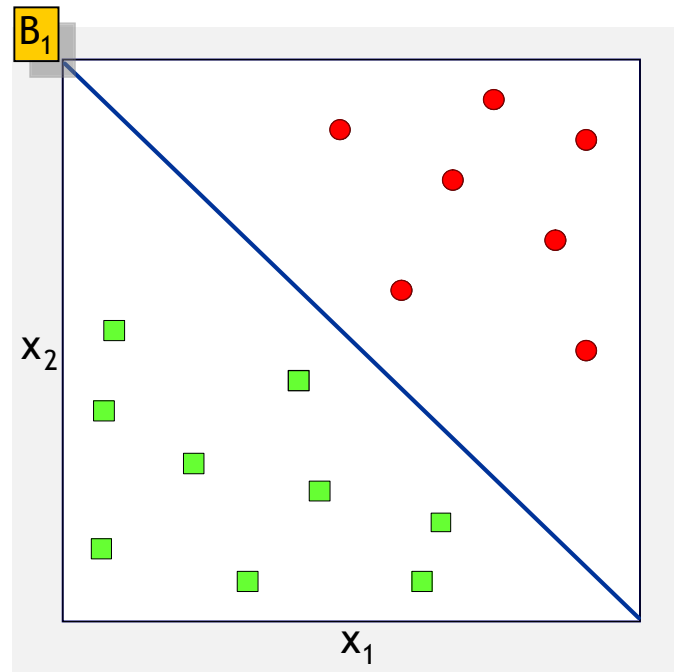
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It is fully described by the following parameters

$$\underline{w} = [w_1, w_2]$$

$$b$$

Is it possible to separate red circles ($\mathbf{y} = +1$) from green squares ($\mathbf{y} = -1$) ?

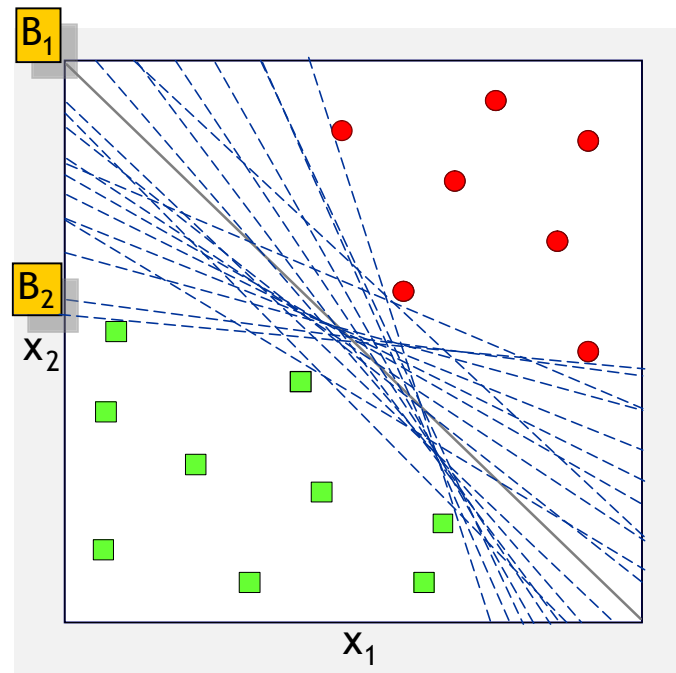
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Multiple lines exist

(LINEAR DECISION BOUNDARIES)

- Are them all equally good?
- Which line should be used for classification?
- Which is the optimal line?



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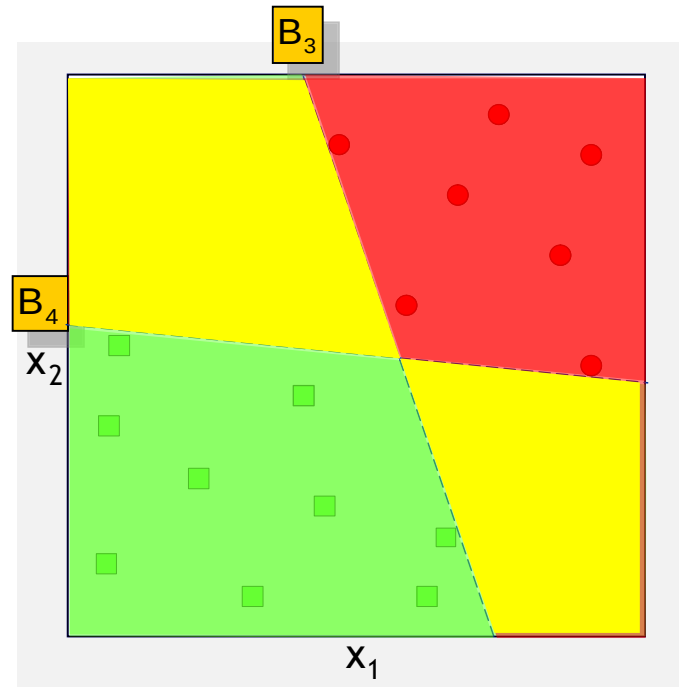
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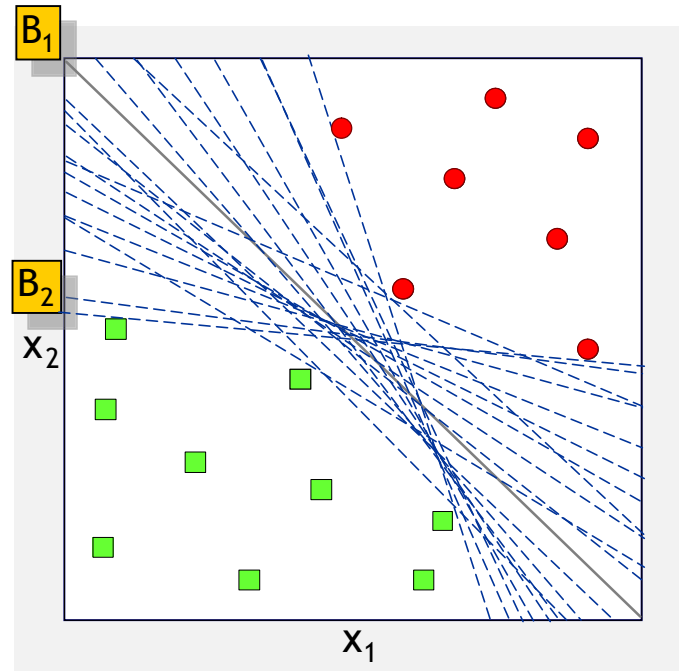
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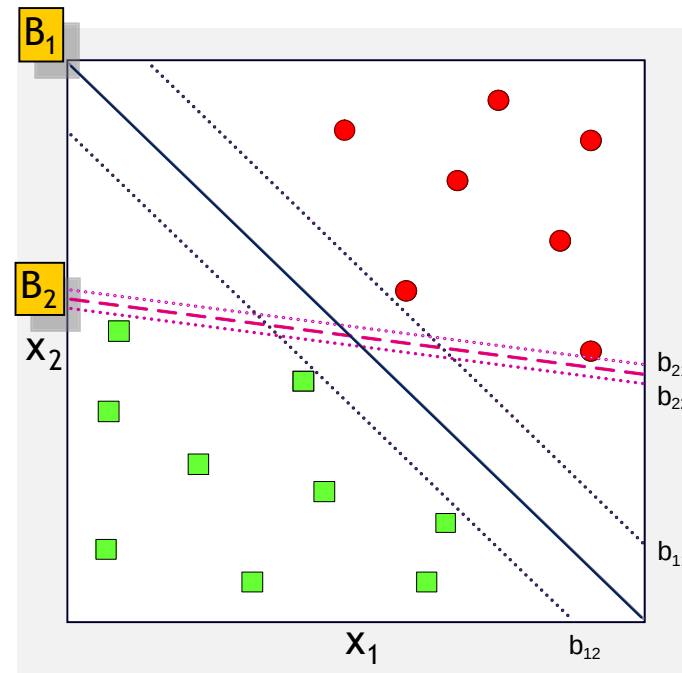
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How to select the **OPTIMAL LINEAR DECISION BOUNDARY**?



How to select the **OPTIMAL LINEAR DECISION BOUNDARY**?

SELECT THE LINE WHICH MAXIMIZES THE MARGIN!!!



B_1 has to be preferred to B_2

Training a **LINEAR HARD-MARGIN SVM** consists of formulating and solving the following mathematical programming problem

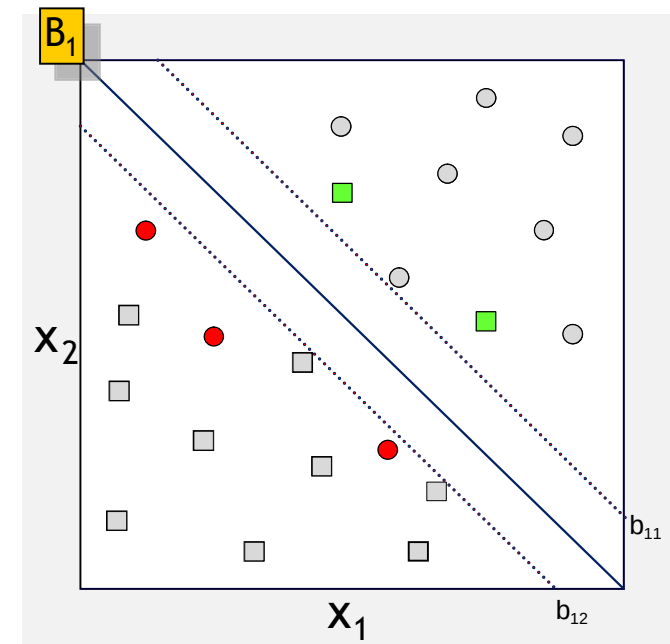
$$\begin{aligned} \min_{\underline{w}, b} \quad & \frac{1}{2} \underline{w} \cdot \underline{w}^T \\ \text{s.t.} \quad & y_i (\underline{w} \cdot \underline{x}_i + b) \geq 1 \quad \forall i = 1, \dots, m \end{aligned}$$

It is a **QUADRATIC PROGRAMMING PROBLEM WITH LINEAR CONSTRAINTS** which must be solved with special numerical techniques.

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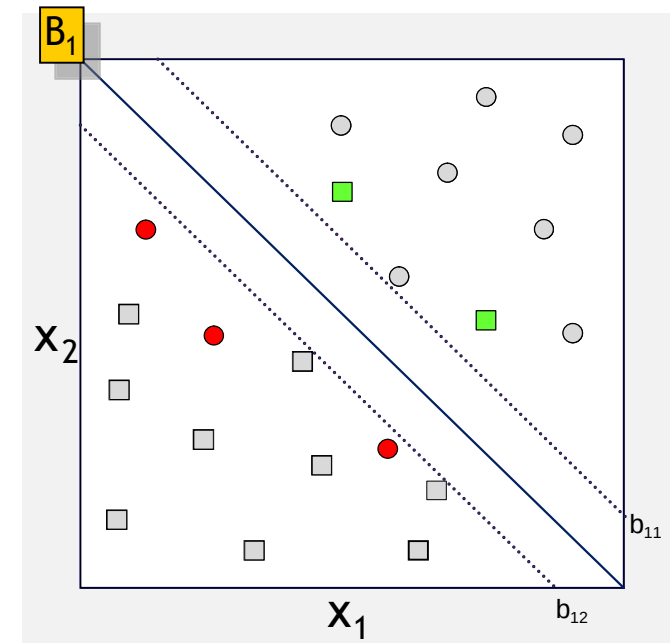


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The **LINEAR HARD-MARGIN SVM** formulation is infeasible and thus we need to introduce the **LINEAR SOFT-MARGIN SVM** formulation.



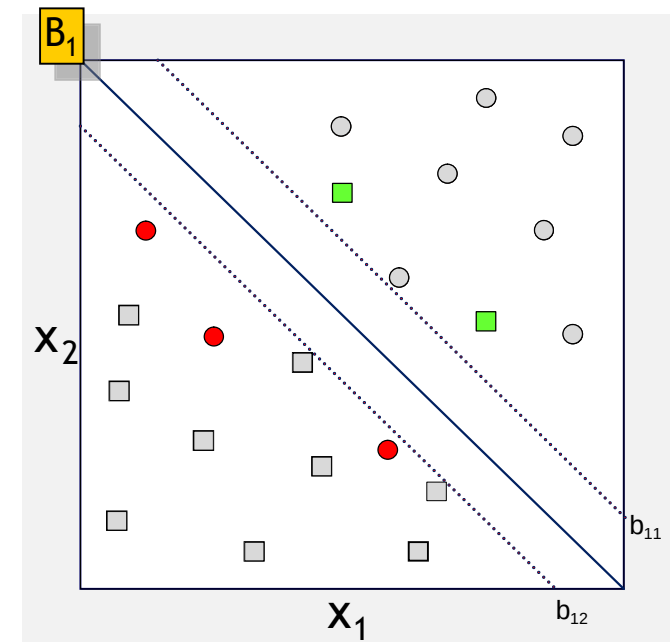
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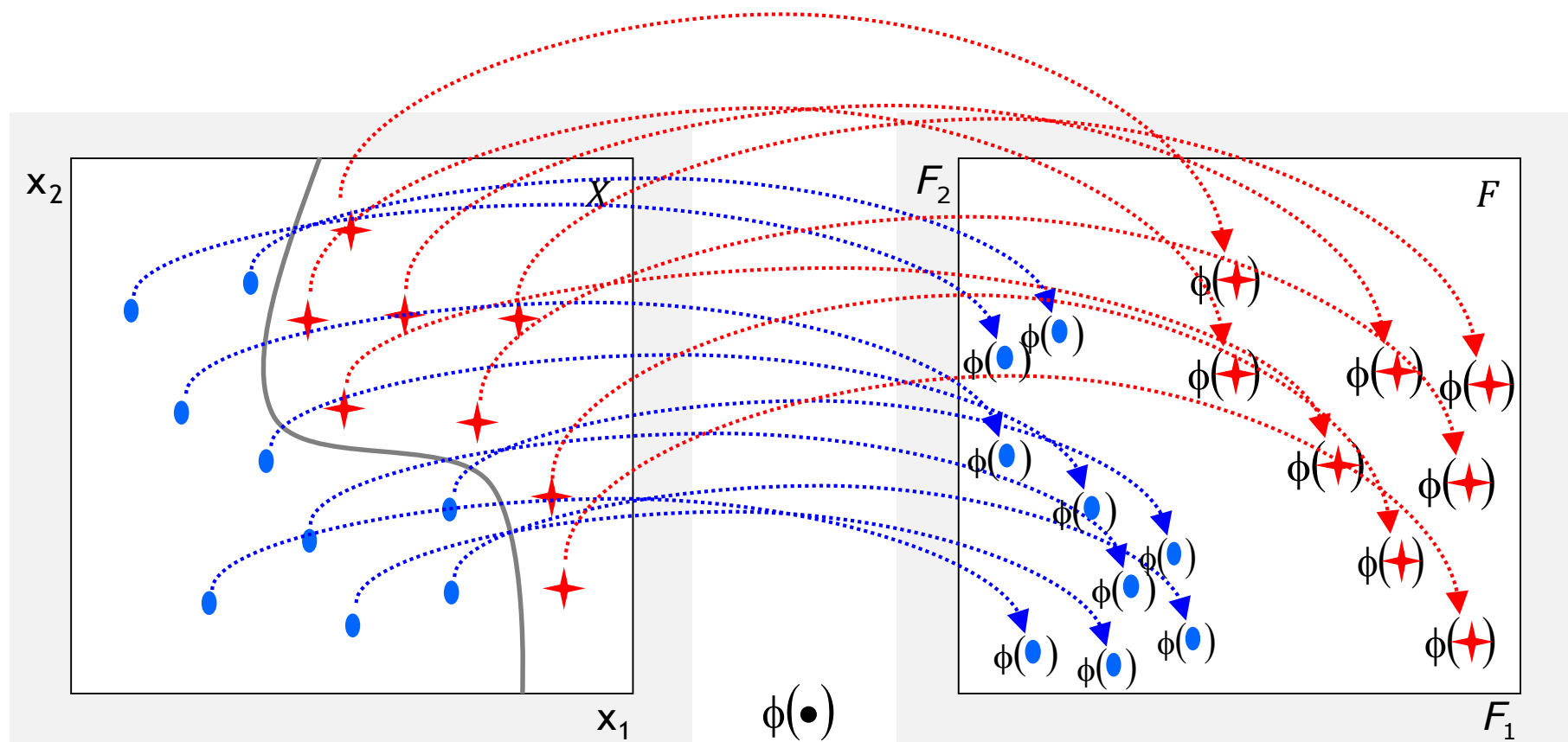
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$$\begin{aligned} \min_{\underline{w}, b, \xi} \quad & \frac{1}{2} \underline{w} \cdot \underline{w}^T + \Delta \sum_{i=1}^m \xi_i \\ \text{s.t.} \quad & \forall_{i=1}^m : y_i(\underline{w} \cdot \underline{x}_i + b) \geq 1 - \xi_i \\ & \forall_{i=1}^m : \xi_i \geq 0 \end{aligned}$$



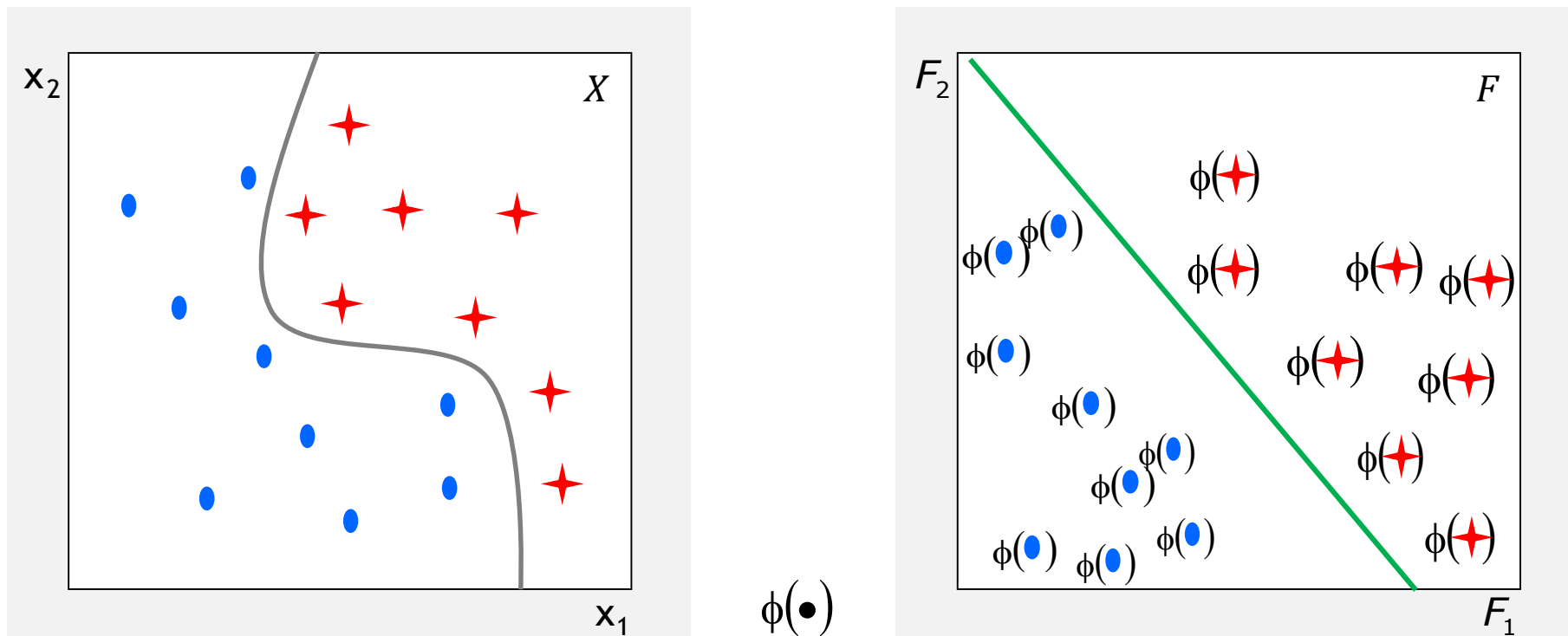


Any tricks to improve SVM when **NON-LINEARLY SEPARABLE DATA SET MUST BE CLASSIFIED?**

Generate a **FEATURE SPACE** where the classification problem becomes easier!!!

Find a **TRANSFORMATION** $\phi(\bullet)$ which **MAPS X TO F** .

The Data projected from the **ORIGINAL SPACE X** to the **FEATURE SPACE F** is **LINEARLY SEPARABLE**

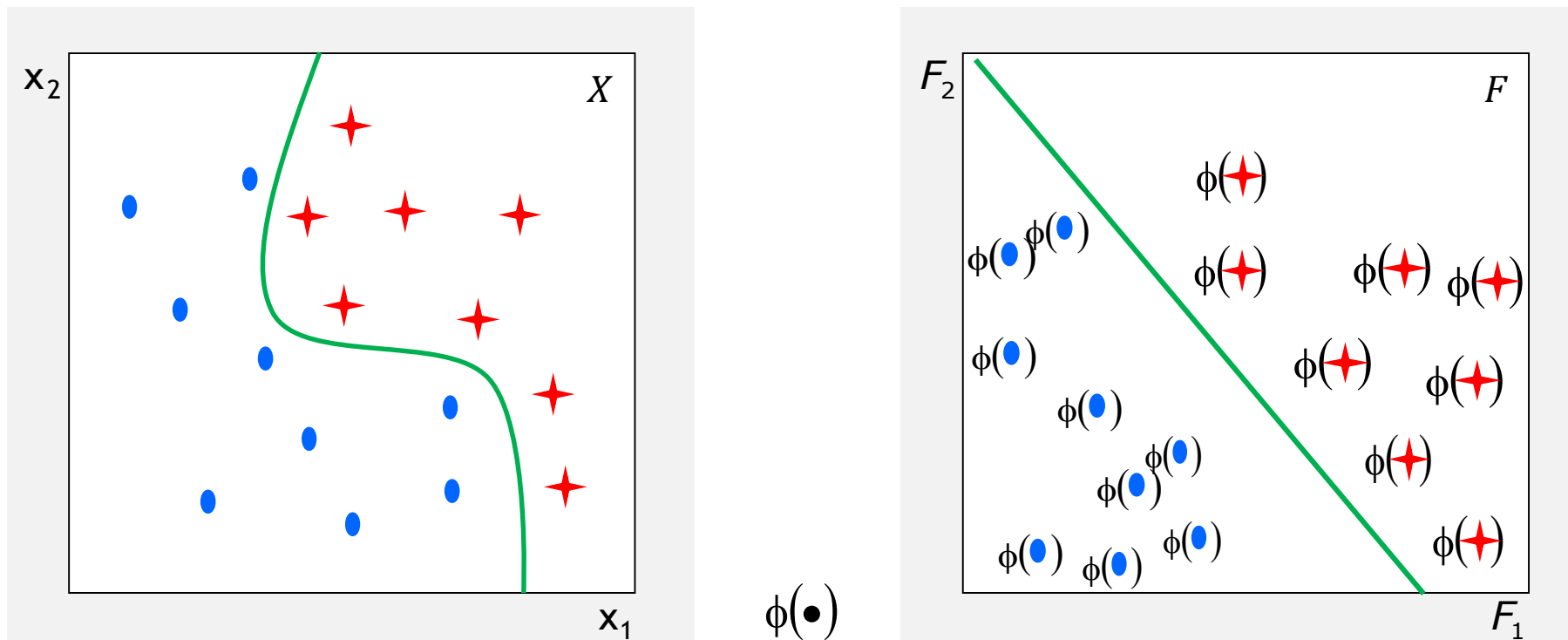


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The **LINEAR DECISION BOUNDARY** in the **FEATURE SPACE F** has the following equation

$$\underline{w} \cdot \phi(\underline{x}) + b = 0$$

Training a **NON-LINEAR SVM** consists of formulating and solving the following mathematical programming problem

$$\begin{aligned} \min_{\underline{w}, b} \quad & \frac{1}{2} \underline{w} \cdot \underline{w}^T \\ \text{s.t.} \quad & y_i (\underline{w} \cdot \phi(\underline{x}_i) + b) \geq 1 \quad \forall i = 1, \dots, m \end{aligned}$$

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The main difference with Linear Soft-margin SVM learning is that instead of using the attributes **X** the **transformed attributes $\phi(\underline{X})$** are used.

The learning algorithm uses the following function

$$K(\underline{u}, \underline{v}) = \phi(\underline{u}) \cdot \phi(\underline{v})$$

K is a similarity function computed in the original attribute space **X** and it is referred to as the **KERNEL FUNCTION**.